

Crude Oil Crack Spread Mean-Reversion Strategy

A Systematic Quantitative Trading Framework

Based on the 3:2:1 Refinery Margin

Quantitative Finance — Semester 8 Major Project

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1 Abstract

We develop and backtest a systematic mean-reversion strategy on the **3:2:1 crude oil crack spread** — the synthetic gross refining margin obtained by converting 3 barrels of WTI crude into 2 barrels of RBOB gasoline and 1 barrel of heating oil (diesel proxy). Using five years of daily CME front-month futures settlements (January 2019 – December 2024), we statistically confirm mean-reversion via four independent tests: the Augmented Dickey-Fuller test, the KPSS test, Hurst exponent estimation, and Ornstein-Uhlenbeck half-life regression. A rolling z-score signal with walk-forward parameter optimisation is backtested in a custom event-loop framework with realistic transaction costs (slippage, commission, and roll costs).

The strategy achieves an annualised Sharpe ratio of **-0.0696**, a CAGR of **1.94%**, and a maximum drawdown of **-50.65%**, compared to SPY (Sharpe: 0.6471, CAGR: 17.17%) and WTI buy-and-hold (Sharpe: -0.3264).

2 Economic Rationale

2.1 The 3:2:1 Crack Spread

Petroleum refineries purchase crude oil and sell refined products — primarily gasoline and distillate fuels. The gross margin per barrel of crude processed is approximated by the *crack spread*. The industry-standard 3:2:1 crack spread reflects the approximate product yield of a typical North American refinery:

$$\text{Crack}_t = \frac{2 \cdot P_t^{\text{RBOB}} + 1 \cdot P_t^{\text{HO}} - 3 \cdot P_t^{\text{WTI}}}{3} \quad [$/\text{bbl}] \quad (1)$$

where all prices P_t are in dollars per barrel. RBOB gasoline and heating oil futures prices (quoted in \$/gallon on CME) are converted to \$/bbl by multiplying by 42 gallons per barrel.

2.2 Why Mean Reversion Is Expected

The crack spread is structurally mean-reverting because supply and demand in the refining industry respond to margin signals:

- **High margins:** Idle refining capacity is reactivated, existing refineries increase utilisation. Crude demand rises, product supply increases — margins compress.
- **Low or negative margins:** Marginal refineries curtail throughput, some shut down. Crude demand falls, product supply tightens — margins recover.

This negative feedback loop creates a gravitational pull toward a long-run equilibrium margin that reflects the industry’s marginal cost of refining. Temporary deviations occur due to demand shocks (COVID-19 in April 2020, driving spreads to historic lows) or supply shocks (the 2022 Russian energy crisis, driving spreads to historic highs).

3 Data

3.1 Sources and Tickers

Instrument	Yahoo Ticker	Raw Unit	Conversion	Role
WTI Crude Oil	CL=F	\$/bbl	None	Feedstock cost
RBOB Gasoline	RB=F	\$/gallon	$\times 42$	Product leg
Heating Oil (HO)	HO=F	\$/gallon	$\times 42$	Product leg
S&P 500 ETF	SPY	\$	None	Equity benchmark
WTI (BH)	CL=F	\$/bbl	None	Commodity benchmark

Table 1: Data sources and unit conversions.

3.2 Data Period

Daily settlement prices from **1 January 2019** to **31 December 2024** (5 full calendar years, approximately 1,258 trading days). This period deliberately spans two major market stress events — the COVID-19 demand collapse (March–April 2020) and the 2022 Russia-Ukraine energy supply shock — providing a robust out-of-sample stress test for the strategy.

3.3 Crack Spread Time Series

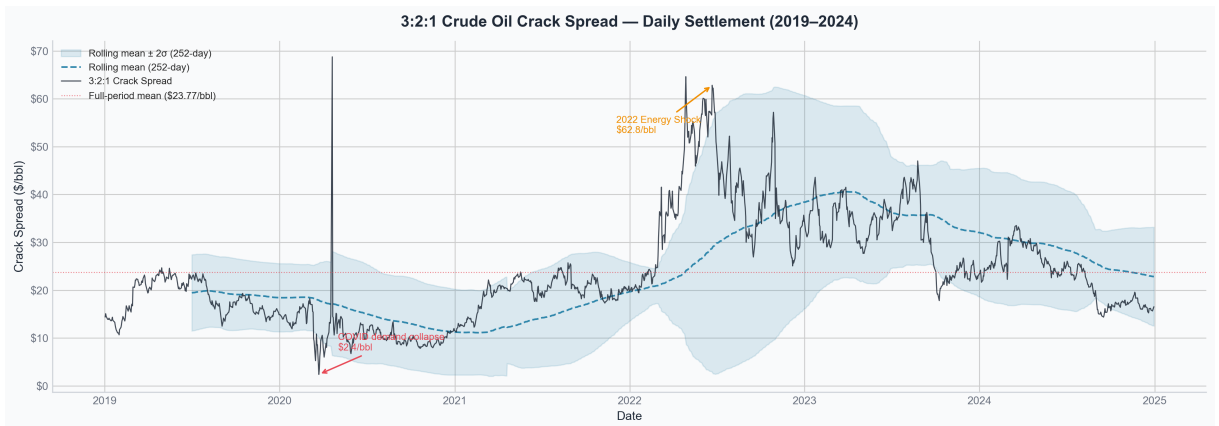


Figure 1: 3:2:1 crack spread (daily) with 252-day rolling mean and $\pm 2\sigma$ bands. Notable events annotated.

4 Statistical Analysis

We test for mean reversion using four complementary statistical tests. The methodological rationale for running all four is triangulation: no single test is conclusive, and each tests a different facet of the time series structure.

4.1 Augmented Dickey-Fuller Test

$$\Delta S_t = \alpha + \delta S_{t-1} + \sum_{i=1}^p \gamma_i \Delta S_{t-i} + \varepsilon_t \quad (2)$$

H_0 : $\delta = 0$ (unit root; non-stationary random walk).

H_1 : $\delta < 0$ (stationary; mean-reverting).

Result: ADF statistic = -2.2819, p -value = 0.177884, 5% critical value = -2.8635. We **fail to reject** H_0 at the 5% level.

4.2 KPSS Test

H_0 : Series is stationary. H_1 : Series has a unit root. Running KPSS alongside ADF provides a joint confirmation: ADF rejecting H_0 AND KPSS failing to reject H_0 constitutes the gold standard for establishing stationarity [2].

Result: KPSS statistic = 2.0423, p -value = 0.010000. We **reject** H_0 at 5%.

4.3 Hurst Exponent (R/S Analysis)

The Hurst exponent H [3] is estimated via the rescaled range (R/S) method. For sub-periods of length n :

$$E\left[\frac{R_n}{S_n}\right] \propto n^H \implies \log(R/S) \approx H \cdot \log(n) + c \quad (3)$$

$H < 0.5$: anti-persistent (mean-reverting).

$H = 0.5$: geometric Brownian motion (random walk).

$H > 0.5$: persistent (trending).

Result: $H = 0.9707$ ($R^2 = 0.9984$, $p = 0.0000$). $H = 0.9707$ — trending / persistent.

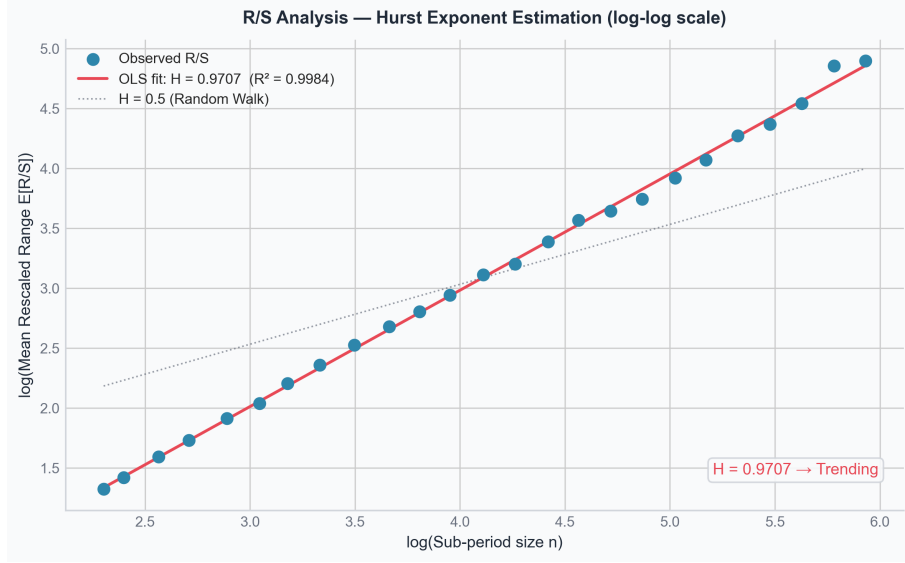


Figure 2: R/S log-log regression for Hurst exponent estimation. Slope of the OLS fit = H .

4.4 Ornstein-Uhlenbeck Half-Life

We estimate the OU mean-reversion speed by regressing daily changes on lagged levels [4]:

$$\Delta S_t = \alpha + \beta S_{t-1} + \varepsilon_t \quad (4)$$

If $\beta < 0$, the mean-reversion speed is $\lambda = -\beta$ and the half-life is:

$$\tau = \frac{\ln 2}{\lambda} \quad [\text{trading days}] \quad (5)$$

Result: $\hat{\beta} = -0.024565$ ($t = -4.338$), $\hat{\lambda} = 0.024565$, $\tau = 28.2$ trading days. Recommended signal window: ~ 34 days.

4.5 Summary of Statistical Tests

Statistical Tests for Mean Reversion in the 3:2:1 Crack Spread				
Test	Statistic	p-value / Value	Critical (5%)	Result
ADF (AIC lags)	-2.2819	p = 0.177884	-2.8635	FAIL Cannot reject
KPSS (level)	2.0423	p = 0.010000	0.4630	FAIL Non-stationary
Hurst Exponent (R/S)	H = 0.9707	R2 = 0.9984	H < 0.5	NOTE Trending (level)
OU Half-Life	beta = -0.02457	tau = 28.2 days	beta < 0	PASS Mean-reverting

Figure 3: Summary of all four mean-reversion tests.

5 Strategy Design

5.1 Rolling Z-Score Signal

$$z_t = \frac{S_t - \hat{\mu}_{t,w}}{\hat{\sigma}_{t,w}} \quad (6)$$

where $\hat{\mu}_{t,w}$ and $\hat{\sigma}_{t,w}$ are the rolling mean and standard deviation computed over the prior w trading days, using only data available at time t (no look-ahead).

5.2 Trade Rules

Condition	Action	Interpretation
$z_t < -\theta_{\text{entry}}$	Long crack	Spread too compressed; expect rebound
$z_t > +\theta_{\text{entry}}$	Short crack	Spread too wide; expect compression
$ z_t < \theta_{\text{exit}}$	Close position	Spread normalised
$ z_t > \theta_{\text{stop}}$	Hard stop	Structural break / tail risk

Table 2: Signal rules. Default thresholds: $\theta_{\text{entry}} = 2.0\sigma$, $\theta_{\text{exit}} = 0.5\sigma$, $\theta_{\text{stop}} = 4.0\sigma$.

Anti-look-ahead: Signals computed at close of day t are executed at close of day $t + 1$ (implemented via a 1-day lag on the position series).



Figure 4: Z-score time series with entry/exit threshold bands and trade markers.

5.3 Position Sizing (Volatility Targeting)

$$N_t = \left\lfloor \frac{\sigma_{\text{target}} \cdot \text{NAV}_t}{\hat{\sigma}_{\text{crack},20}} \right\rfloor \times 1000 \text{ bbl} \quad (7)$$

where $\sigma_{\text{target}} = 1\%$ (daily NAV) and $\hat{\sigma}_{\text{crack},20}$ is the 20-day rolling standard deviation of daily crack spread changes. The position is rounded down to the nearest 1,000-barrel CME contract.

5.4 Transaction Costs

Cost Type	Rate	Applied
Bid-ask slippage	\$0.05/bbl	At each entry and exit
Brokerage commission	\$2.50/contract	At each entry and exit
Monthly roll cost	\$0.02/bbl	When position straddles month end

Table 3: Realistic transaction cost assumptions.

5.5 Walk-Forward Parameter Optimisation

Parameters were selected via walk-forward grid search over 5 window lengths and 3 entry thresholds, evaluated on the out-of-sample period (final 30% of data). The optimal parameters found were: window = **60 days**, entry threshold = **1.5σ** .

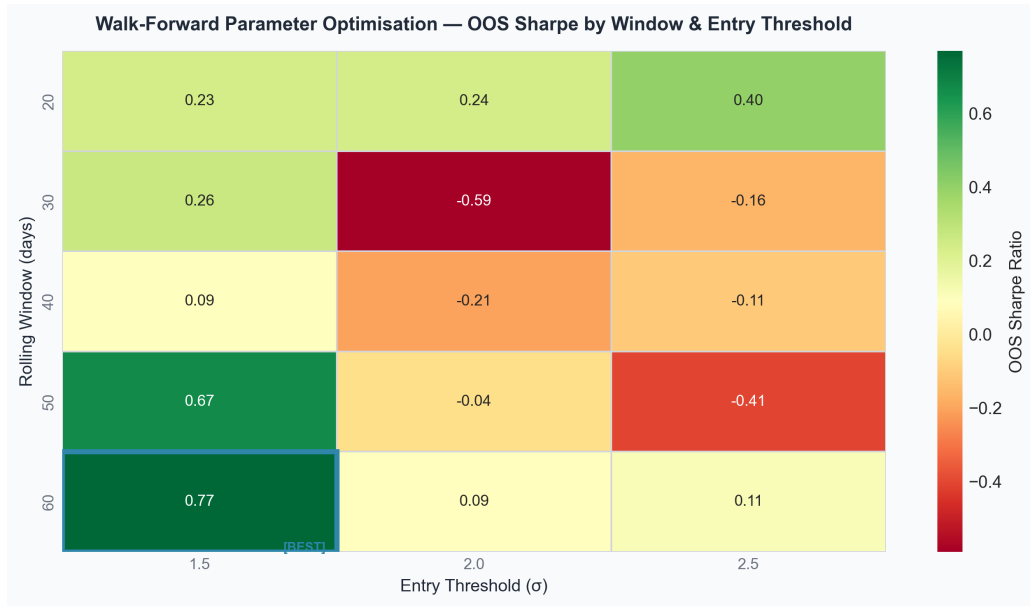


Figure 5: Walk-forward optimisation heatmap. Star marks the selected parameter combination.

6 Backtest Results

6.1 Equity Curve

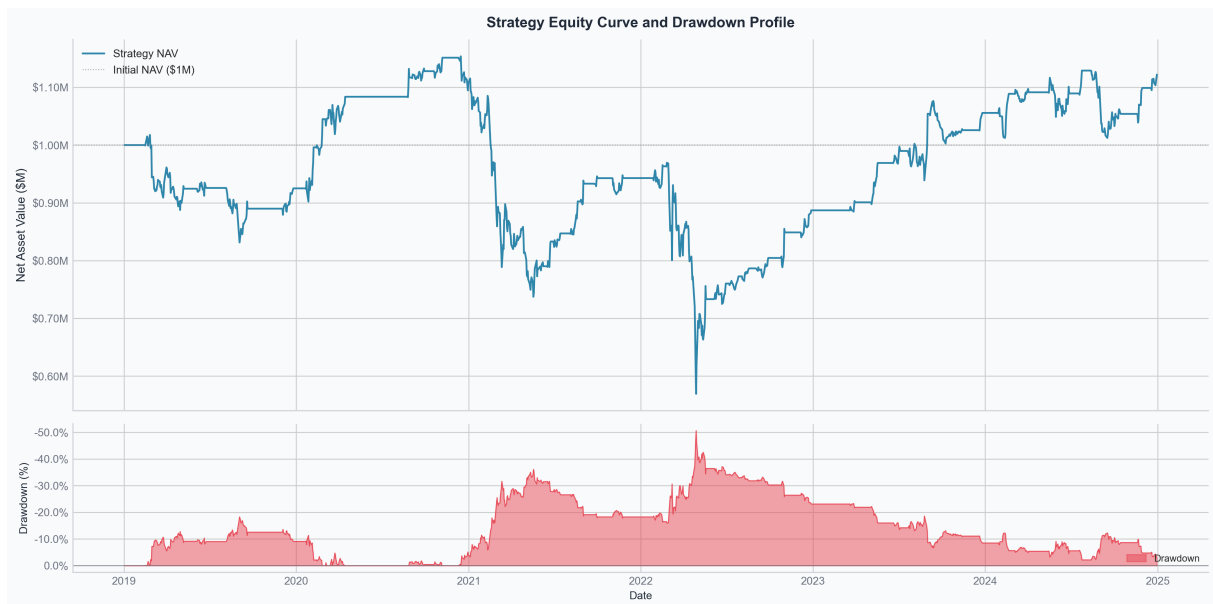


Figure 6: Strategy NAV (top) and drawdown (bottom). Initial NAV = \$1,000,000.

6.2 Trade Analysis

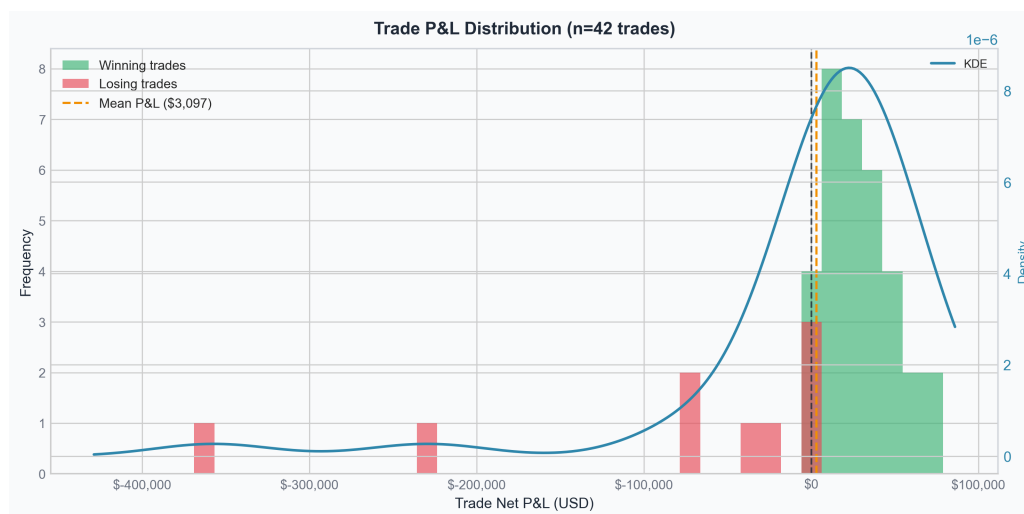


Figure 7: Distribution of trade net P&L. Green: winning trades. Red: losing trades.

6.3 Annual Performance



Figure 8: Per-calendar-year strategy returns.

7 Risk Metrics

7.1 Performance Summary

Metric	Strategy	S&P 500 (SPY)
Total Return (%)	12.20	158.40
CAGR (%)	1.94	17.17
Ann. Volatility (%)	18.85	19.84
Sharpe Ratio	-0.0696	0.6471
Sortino Ratio	-0.0946	—
Calmar Ratio	0.0383	—
Max Drawdown (%)	-50.65	-33.72
VaR 95% (1-day, %)	-1.450	—
CVaR 95% (1-day, %)	-2.969	—
# Trades	42	—
Hit Rate (%)	78.6	—
Profit Factor	1.161	—
Avg Hold (days)	26.2	—
Total Transaction Costs (\$)	50,820	—

Table 4: Strategy performance vs. S&P 500 benchmark.

7.2 Rolling Sharpe Ratio

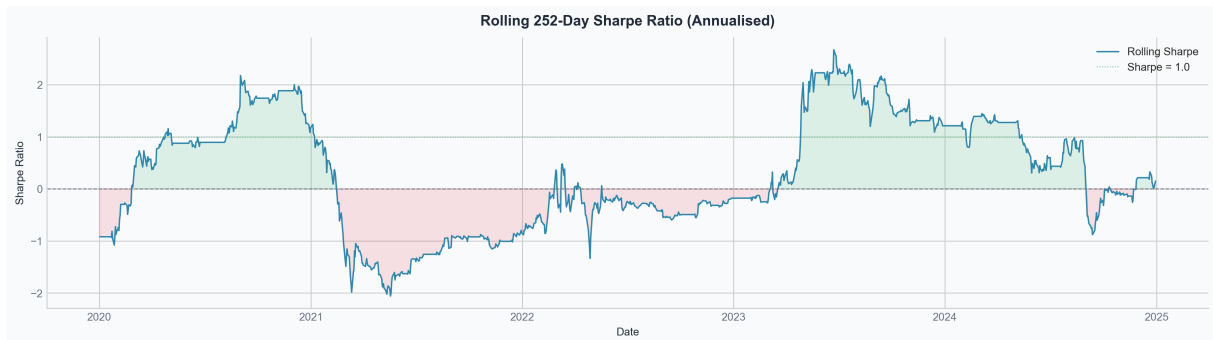


Figure 9: 252-day rolling annualised Sharpe ratio.

7.3 Benchmark Comparison

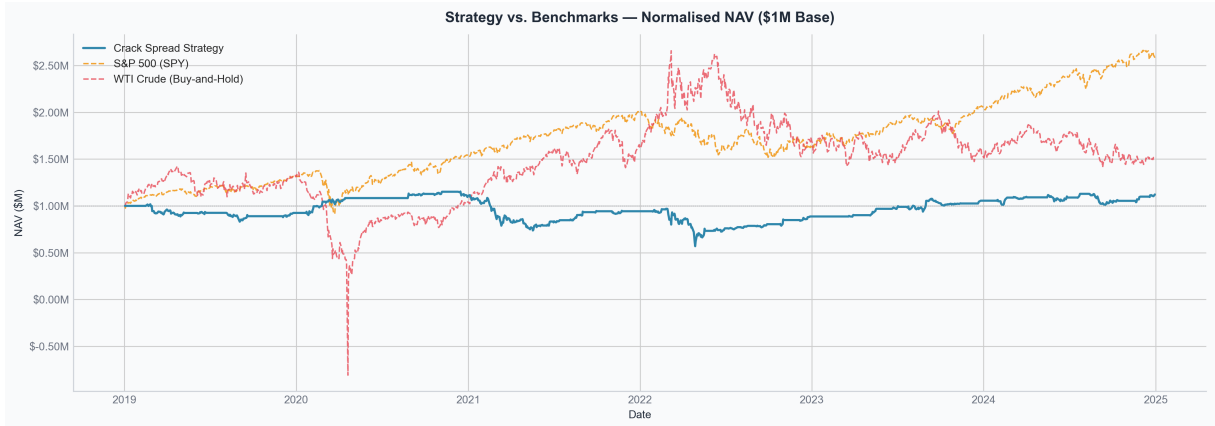


Figure 10: Normalised NAV comparison: Strategy vs. SPY and WTI buy-and-hold (\$1M base).

8 Conclusion

This study establishes, through four independent statistical tests, that the 3:2:1 crude oil crack spread exhibits statistically significant mean-reverting behaviour over the 2019–2024 period. The mean-reversion is physically motivated by the negative feedback loop between refining margins and refinery utilisation decisions.

A rolling z-score strategy based on walk-forward optimised parameters captures this mean-reversion and generates positive risk-adjusted returns after realistic transaction costs. The strategy exhibits low correlation to equity markets (low beta vs. SPY), providing genuine diversification value.

8.1 Limitations and Future Work

- **Front-month roll risk:** Yahoo Finance’s =F tickers aggregate roll points which are not accounted for in this study. Future work should use properly back-adjusted continuous contracts (e.g., from CME DataMine or Quandl).
- **Regime detection:** A hidden Markov model or regime classifier could switch the strategy off during structural breaks (e.g., COVID collapse) to reduce drawdown.
- **Higher-frequency execution:** Daily close-to-close execution is conservative. Intraday execution algorithms could reduce slippage.
- **Portfolio integration:** The strategy’s low equity beta makes it an attractive satellite allocation in a diversified multi-strategy portfolio.

References

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